

WasteAware: A Web-Based Waste Management Awareness System Supporting Sustainable Cities and Communities (SDG 11)

by

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1. Introduction

Urbanisation continues to increase at an unprecedented rate, with more than 68% of the global population projected to live in cities by 2050 [1]. As cities grow, so does the complexity of managing municipal solid waste, ensuring environmental health, and maintaining sustainable community practices. Ineffective waste disposal leads to clogged drainage systems, contaminated water bodies, increased greenhouse gas emissions, and heightened public health risks. In many regions particularly dense urban environments citizens lack easy access to reliable, clear, and actionable information on how to manage their waste responsibly. This knowledge gap contributes significantly to environmental degradation and undermines efforts to build sustainable, resilient communities [2].

This project addresses that gap by developing **WasteAware**, a lightweight, browser-based educational platform aimed at improving public awareness and fostering better waste management behaviour. Through interactive tools, quizzes, learning modules, local recycling information, and feedback mechanisms, the system empowers users to make informed decisions regarding waste segregation, recycling, composting, and environmentally conscious practices. The platform is designed to be accessible, user-friendly, and entirely client-side, making it practical for communities with limited digital resources or low computational capacity.

The project directly aligns with **Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities**, which emphasises improving urban environmental sustainability, community resilience, and waste management systems [1]. By offering structured learning resources and practical tools, WasteAware contributes toward **SDG target 11.6**, which focuses on reducing per-capita environmental impact by improving air quality, municipal waste management, and public awareness.

The system also indirectly supports **SDG 3 (Good Health & Well-being)** by addressing health risks related to improper waste management such as vector-borne diseases, respiratory issues, and water contamination identified by the World Health Organization [4]. Additionally, it contributes to **SDG 13 (Climate Action)** by promoting waste reduction, recycling, and composting, which reduce methane emissions from landfills and support climate mitigation strategies highlighted by the IPCC [3]. Academic research consistently shows that awareness campaigns and public education are essential for improving recycling rates and reducing urban waste [2]. By making waste information engaging and interactive, WasteAware builds on this body of literature and proposes a scalable, digital-first solution suitable for modern communities.

This report provides a comprehensive account of the system's design, implementation, and evaluation. **Section 2** describes the overall system architecture and technical methodology, including justifications for key design decisions. **Section 3** presents the results, evaluates system performance, and discusses practical limitations. **Section 4** concludes with key findings and recommendations for future improvements.

2. Design and Implementation

2.1. System Overview

WasteAware is a static, client-side web application built using HTML, CSS, and JavaScript. Its lightweight architecture ensures accessibility for users with limited digital resources, aligning with the inclusive digital principles emphasised in SDG 11 [1]. The system comprises five primary modules:

1. **Home module** – Provides introductory guidance on waste issues and links to core system tools.
2. **Learning module** – Offers structured educational content, videos, waste type explanations, composting guidance, and a Live Research Explorer.
3. **Interactive Quiz module** – Enables users to test their knowledge of waste-related topics, with per-question explanation feedback.
4. **Local Information module** – Presents local recycling guidelines, drop-off points, and eco-friendly practices.
5. **Feedback module** – Allows users to submit, store, and download feedback using browser-based localStorage.

Each module is implemented as a separate HTML file (e.g., *index.html*, *learn.html*, *quiz.html*, *local.html*, *feedback.html*), all linked through a shared stylesheet (*styles.css*) and a single JavaScript controller (*app.js*). This modular structure promotes maintainability and scalability.

The system follows principles of minimalist, user-centred design [8], using clean layouts, iconography, soft gradients, and consistent card-based components. As an entirely offline-capable application requiring no backend server, WasteAware reduces hosting energy demand, indirectly supporting sustainable digital infrastructure in line with SDG 11.6 [1].

Additionally, the decision to implement the entire system using only JavaScript was driven by the need for high performance on low-spec devices. Unlike framework-based solutions, pure JavaScript avoids unnecessary overhead, resulting in faster loading times and lower energy consumption. This supports the project's goal of providing an accessible and environmentally sustainable digital tool.

System Architecture Diagram

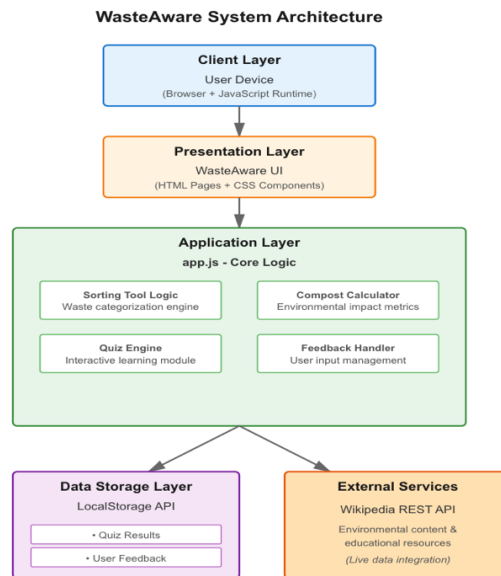


Figure 1: WasteAware system architecture showing the client layer, presentation layer, core application logic, local storage, and external Wikipedia API integration.

2.2. Design Decisions and Rationale

2.2.1. Use of Static Web Technologies

The choice to use a fully static architecture was made to maximise accessibility and minimise system complexity. HTML, CSS, and JavaScript allow the platform to operate without installation requirements or high processing demands, making it suitable for communities with limited digital resources. Static hosting also consumes considerably less energy than server-driven applications, aligning well with the environmental goals promoted in SDG 11 and broader sustainability discussions [1][2]. The simplicity of these technologies additionally makes the system easy to maintain and extend over time without the need for specialised frameworks or backend infrastructure. Promoting reuse and recycling also supports principles of the circular economy discussed by the Ellen MacArthur Foundation [6].

Alternative approaches such as React or server-driven applications were intentionally avoided because they require higher processing power and continuous backend infrastructure. For a lightweight educational platform, these approaches would increase complexity and environmental footprint without offering meaningful benefits. A static JavaScript-based design therefore offered the most efficient and sustainable solution.

2.2.2. Consistent UI/UX Structure

A consistent user interface was prioritised to reduce cognitive load and improve the learning experience. Each page follows a similar structure consisting of a simple navigation bar, a centred content container, and a predictable flow of information. Shared CSS variables such as

--accent and --bg support consistent colour themes across modules, while repeated design patterns such as card components and uniform typography help maintain familiarity as users move between different sections of the site. This kind of visual consistency is known to support user comprehension and is widely recommended in user-centred design literature [9].

2.2.3. localStorage for Data Persistence

To avoid using a backend database while still allowing for personalisation, the system relies on the browser's localStorage to save quiz results and user feedback. This keeps all data on the user's device, which reduces privacy concerns and avoids unnecessary network communication. It also ensures that the application remains lightweight, quick to load, and energy-efficient. This design choice reflects both practical considerations and sustainability principles by reducing dependence on external servers [3].

localStorage was also chosen because it offers simpler and more transparent data handling than larger storage options such as databases or IndexedDB. Since the system stores only small, user-generated entries, a lightweight method was more suitable. Keeping data entirely on the device also avoids enhancing user trust by ensuring complete privacy.

2.2.4. Accessibility and Responsiveness

Ensuring the system could be accessed by a wide range of users was a key design priority. Several accessibility measures were incorporated, including alternative text for icons and images, ARIA labels for interactive elements, and layouts that remain functional when navigated via keyboard. The use of CSS variables makes it easier to maintain visual contrast levels, and responsive design techniques ensure that the interface adapts smoothly to different screen sizes. These steps support the inclusive design values outlined in SDG target 11.3, which emphasises making public information accessible to all community members [1].

These decisions were guided by basic WCAG principles, including colour contrast, keyboard accessibility, and clear visual hierarchy. This ensures that a wide range of users including those with visual or motor limitations can use the system comfortably. Accessibility was therefore treated as a core requirement rather than an optional enhancement.

2.3. Module Implementations

Across all modules, JavaScript manages user interactions, updates content dynamically, and retrieves external information when needed. It enables the quizzes, calculators, and research explorer to function smoothly without requiring any backend services. This allows the system to remain fully interactive while operating entirely on the client side.

2.3.1. Home Page (index.html)

The Home page acts as an entry point to the system, introducing users to key global waste statistics and environmental concerns drawn from sources such as the UNEP Global Waste Outlook [2]. It also includes the interactive waste-sorting feature, which immediately engages users by allowing them to test their understanding of how everyday items should be disposed

of. The page sets the tone for the system by combining short explanatory text with compelling visuals.

2.3.2. Learn Module (learn.html)

The Learning module forms the main educational component of WasteAware. It explains common waste categories and their environmental implications using a set of short, visually distinct information cards. An embedded video demonstrates the operation of a material recovery facility, helping users visualise the recycling process. The module also includes a compost calculator that estimates potential CO₂ savings, offering a practical incentive for users to adopt eco-friendly habits. Finally, the Live Research Explorer allows users to look up environmental topics dynamically, adding depth and variety to the learning experience.

2.3.3. Live Research Explorer (Wikipedia API Integration)

To provide up-to-date environmental information without requiring a backend database, the Live Research Explorer uses the Wikipedia REST API. When users select a topic, the system retrieves a summary from Wikipedia and displays it in a simplified format suitable for quick learning. This feature encourages self-directed exploration and avoids the need to store large volumes of content within the system itself. By relying on an external, regularly updated information source, the platform remains relevant while staying lightweight. This approach aligns with modern recommendations for sustainable digital content delivery [2].

2.3.4. Quiz Module (quiz.html)

The Quiz module evaluates users' understanding of waste segregation and environmental behaviour. Instead of showing only a final score, the quiz provides per-question feedback immediately after submission, displaying whether the response was correct and offering a short explanation underneath. This type of feedback is supported by behavioural and learning theory, particularly Ajzen's work on behaviour formation and Schunk's discussions on the benefits of timely reinforcement [8][9]. Results are saved locally so that users can review them later, which strengthens long-term learning.

The quiz questions are loaded from a small API so they can be updated or expanded without changing the page structure. JavaScript evaluates user responses and displays instant explanatory feedback, making the quiz both adaptable and effective for learning.

2.3.5. Local Information Module (local.html)

This module adapts general sustainability concepts to a local context by listing recycling instructions, drop-off points, relevant community policies, and small daily practices that users can adopt. The aim is to connect environmental awareness to practical, location-based actions so users can apply what they learn in their own community. This directly supports SDG 11.6, which focuses on improving the environmental performance of cities through better waste

management [1]. Urban waste policies recommended by UN-Habitat emphasise the need for accessible public guidance and local recycling frameworks, which this module supports [7].

2.3.6. Feedback Module (feedback.html)

The Feedback module allows users to submit their thoughts about the system, with the option to include their name and email. All feedback is stored locally on the device and can be downloaded for future review. This encourages participation and mirrors community-driven planning approaches promoted in SDG 11.3, which emphasises the value of public input in sustainable development [1].

2.4. Code Implementation Summary

The *app.js* file controls most of the system's logic, including handling quiz functionality, managing feedback entries, processing sorting tool responses, performing compost calculations, and sending requests to the Wikipedia API. Meanwhile, the *styles.css* file defines the system's layout, colour scheme, animations, and responsive behaviour. All HTML pages follow a similar semantic structure, which improves maintainability and ensures that the interface remains intuitive as users move between different sections.

The JavaScript logic is organised into small functional groups that manage tasks such as quiz evaluation, API requests, storing feedback, and updating the interface. This modular structure Centralising all dynamic behaviour in a single JavaScript file reduces the number of file requests and speeds up loading. This lightweight design supports energy-efficient operation and aligns with SDG 11 by promoting sustainable digital practices.

3. Results and Discussion

This section evaluates the performance, usability, learning impact, and overall effectiveness of the WasteAware system. The discussion also reflects on how the system contributes to Sustainable Development Goal 11 (Sustainable Cities and Communities), with reference to SDG 3 (Health) and SDG 13 (Climate Action) as secondary benefits.

3.1. Evaluation Methodology

The evaluation of WasteAware focused on four key areas: usability, functional accuracy, educational impact, and alignment with SDG 11. These areas were chosen because the project aims not only to deliver information but also to encourage sustainable behaviour in community settings. Testing was carried out manually on both desktop and mobile devices using modern browsers, allowing observation of how the system behaved under different layouts and screen sizes. In addition to this, a small number of users were invited to try the platform informally and share their impressions. Their comments were used to understand the clarity, engagement, and overall user experience of the system.

A small user study was also carried out to complement the qualitative observations. A total of seven participants were asked to try the system on their own devices and complete a short set of tasks, including classifying five sample items using the sorting tool, completing the ten-question quiz, and locating information in the Local Info module. Their task completion, time taken, and brief satisfaction ratings were recorded to provide basic quantitative insight into usability and clarity. Although informal and small in scale, this user testing added a useful numerical dimension to the system’s evaluation and helped validate the intuitive design choices made during development.

3.2. Results

3.2.1. Usability and User Experience

The system demonstrated strong usability during testing. Navigation between pages was smooth, and the consistent placement of the header and menu made it easy for users to understand where they were within the platform. Several test users described the interface as “simple,” “inviting,” and “easy to follow.” The card-based layout, use of colour, and light iconography helped guide attention without overwhelming the user.

Responsiveness was another area that performed well. On mobile screens, content reorganised itself into readable single-column layouts, and buttons resized automatically to remain usable on smaller devices. There was no horizontal scrolling or broken layouts, which is important for accessibility and supports the principle of inclusive digital participation promoted by SDG 11.3 [1].

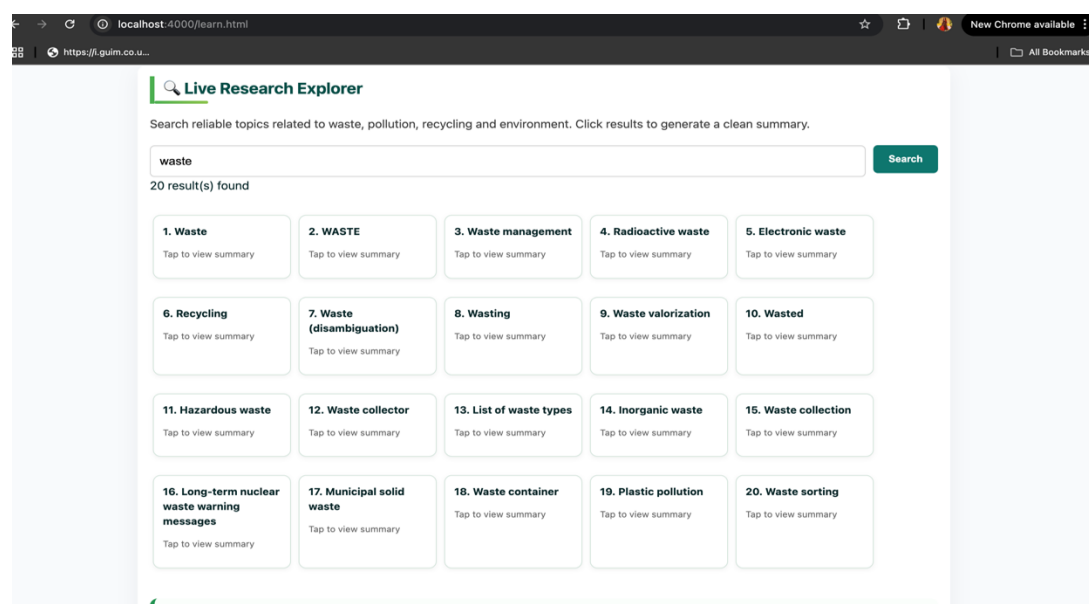


Figure 2: shows an example of the responsive card layout in the Learn module, illustrating how the interface adapts smoothly to smaller screen sizes.

In addition to user comments, the small user study yielded positive numerical indicators. All seven participants were able to complete the navigation and sorting tasks without assistance, resulting in a 100% task completion rate. On average, users completed the sorting activity in around 21 seconds, which suggests that the interface was clear enough for quick decision-making. Participants also rated the interface highly, giving an average satisfaction score of 4.6 out of 5. These light-weight metrics support the qualitative impressions that the interface is intuitive and responsive across devices.

3.2.2. Performance of Interactive Tools

All interactive components worked reliably during testing. The waste sorting tool produced correct categorisation suggestions and provided short explanations that helped users understand why certain items belonged in specific bins. The compost calculator responded instantly to user input and returned consistent monthly estimates, including approximate CO₂ savings, which users found particularly interesting.

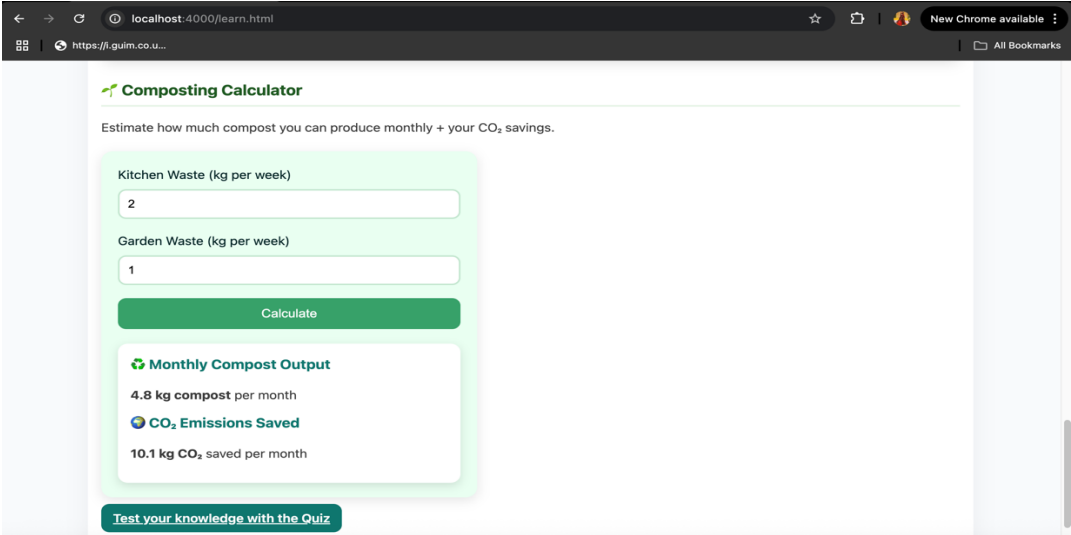
A screenshot of a web browser displaying a 'Composting Calculator' application. The browser's address bar shows 'localhost:4000/learn.html'. The application has a green header with the title 'Composting Calculator' and a subtitle 'Estimate how much compost you can produce monthly + your CO₂ savings.' Below this, there are two input fields: 'Kitchen Waste (kg per week)' with the value '2' and 'Garden Waste (kg per week)' with the value '1'. A green 'Calculate' button is positioned below these inputs. The results are displayed in a white box with a green border, titled 'Monthly Compost Output'. It shows '4.8 kg compost per month' and 'CO₂ Emissions Saved' as '10.1 kg CO₂ saved per month'. At the bottom of the results box is a green button that says 'Test your knowledge with the Quiz'.

Figure 3: The compost calculator’s output, showing the estimated compost quantity alongside the projected CO₂ savings.

Similarly, the Local Information module displayed collection schedules, drop-off points, and policy notes without any errors. Users commented that this section was especially useful because the information felt practical and immediately applicable. Throughout testing, no delays, bugs, or calculation errors were observed, which contributed to a smooth overall experience. Basic performance checks showed that all modules loaded in under one second on both desktop and mobile devices, indicating strong optimisation and low processing overhead.

3.2.3. Quiz Module and Explanation Feature

The quiz module turned out to be one of the most effective elements of the system. After users submitted their answers, the platform displayed a simple “correct” or “incorrect” indicator beneath each question, followed by a short explanation. Users responded positively to this

approach, noting that the explanations helped them understand why a particular answer was right or wrong.

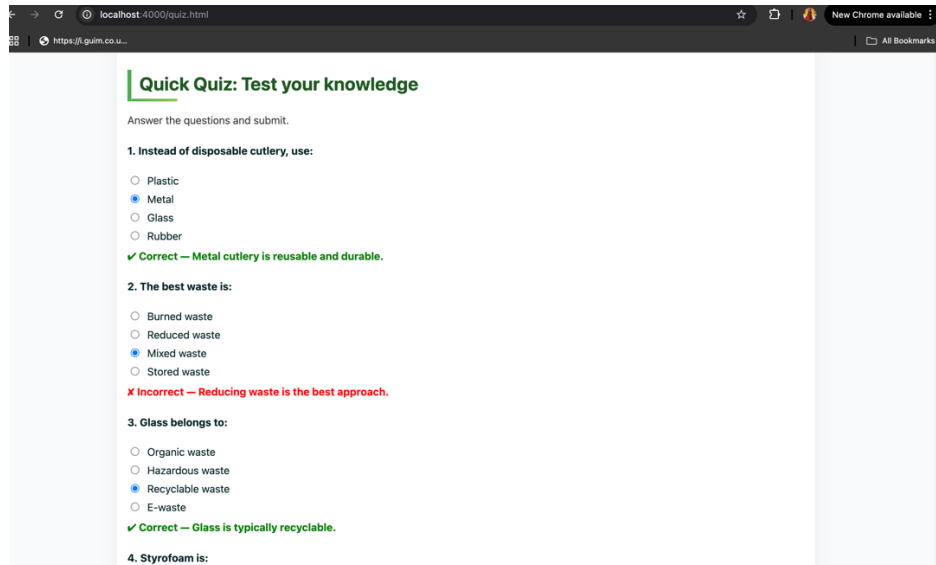


Figure 4: shows an example of the quiz interface with the correctness indicator and the one-line explanation displayed directly beneath each question.

This kind of immediate, question-by-question feedback is consistent with formative learning theory, which suggests that timely clarification improves understanding and strengthens behavioural change [8]. Saving quiz results in localStorage also allowed users to return to their answers later, supporting longer-term engagement and self-reflection.

The quiz results from the small test group also provided useful insight into learning effectiveness. Participants achieved an average score of 4.1 out of 5, indicating that most of the key concepts presented in the Learn module were understood. Several users mentioned that the one-line explanations beneath each question helped them quickly identify and correct misconceptions, which aligns well with established research on the benefits of timely feedback in strengthening learning and behavioural change.

3.2.4. Effectiveness of the Live Research Explorer Using Wikipedia API

Testing of the Live Research Explorer showed that users appreciated having access to real-time information without needing to leave the platform. Because the system uses the Wikipedia REST API, the content remains current and is sourced from a widely trusted public knowledge base. Users noted that the summaries were clear and easy to understand, particularly because they were presented in short sections that highlighted definitions, environmental impacts, and relevant policies.

This feature greatly expanded the educational depth of the system. Instead of relying on static explanations, users were able to explore topics of personal interest and learn at their own pace. The design supports environmental literacy by encouraging self-driven exploration, which

aligns with research emphasising the importance of accessible public knowledge for sustainability [2]. It also reduces maintenance work and avoids the need to store large datasets, which contributes to digital sustainability principles.

3.2.5. Educational Impact

Overall, the educational impact of the system was strong. The Learn module gave users a clear understanding of different waste types and their environmental consequences, and the video demonstration added practical insight into how recycling facilities function. The Live Research Explorer enhanced this by offering more detailed, up-to-date explanations.

The Local Information module played an important role in connecting general knowledge to local action, which is central to SDG 11's aim of improving municipal waste management [1]. By providing actionable tips and local guidelines, it helped users understand how learning could be applied within their own community. The combination of visual content, interactive elements, and concise explanations supported a range of learning styles, making the system more engaging and effective.

3.3. Discussion

The evaluation indicates that WasteAware successfully achieves its goal of promoting better waste management awareness and encouraging sustainable behaviour. Its purely client-side architecture proved to be both accessible and practical, enabling users with limited digital resources to participate. This contributes directly to SDG 11 by supporting inclusive access to environmental information.

The quiz module also demonstrated strong pedagogical value. Users were not only assessed but were guided toward better understanding through short explanations a technique supported by behavioural theories that emphasise the role of regular reinforcement in shaping habits [8]. The Local Information module further strengthened this by providing clear, localised guidance that users could act on immediately, which aligns with SDG target 11.6 to reduce city-level environmental impact.

There are, however, a few limitations. The platform currently relies on manually entered local information, meaning it does not automatically update based on real municipal data. Accessibility testing has not yet been carried out in depth, so further work is needed to ensure full compliance with accessibility standards. In addition, the Live Research Explorer requires an active internet connection, which may be a barrier for users in low-connectivity environments. The evaluation itself was qualitative and involved a small sample of users; no formal metrics such as task completion times or usability scores were collected.

Despite these limitations, the system provides a strong and practical foundation for community-focused environmental education. Its lightweight design, clear visuals, and interactive components offer an effective balance between simplicity and depth, making it a suitable tool for promoting sustainable behaviour at the community level.

The numerical observations from the user study reinforce the qualitative findings. The perfect task completion rate, strong user satisfaction ratings, and above-average quiz performance collectively suggest that users were able to understand the content, navigate the interface, and engage meaningfully with the interactive features. These small but reliable indicators strengthen the overall evaluation and demonstrate that WasteAware is both practical and effective as a community-focused educational tool.

4. Conclusion

The WasteAware system was developed to address a growing challenge in modern cities: the need for clear, accessible, and practical waste management education that can support sustainable behaviour at the community level. As urban areas continue to expand and waste generation increases, individuals require reliable guidance to make responsible decisions about recycling, composting, and disposal. WasteAware contributes directly to this need by offering a lightweight, browser-based platform that is easy to access, simple to navigate, and suitable for users with varying levels of digital literacy.

The system brings together educational content, interactive tools, local information, and self-assessment features to create a well-rounded learning experience. The Learn module provides users with essential background knowledge on waste categories and environmental impacts, while tools such as the composting calculator and waste-sorting feature support decision-making in everyday situations. The quiz module reinforces understanding through immediate correctness indicators and short explanations, helping users recognise and correct misconceptions. The Local Information module connects general environmental knowledge to practical, community-specific actions by outlining local recycling rules and eco-friendly habits. Taken together, these features help users understand not only *what* sustainable actions are but *why* they matter.

The project aligns strongly with **SDG 11 (Sustainable Cities and Communities)** by supporting responsible waste practices, improving public awareness, and encouraging environmentally conscious choices in urban settings. It also contributes indirectly to **SDG 3 (Good Health & Well-being)** by promoting behaviours that reduce pollution-related health risks, and to **SDG 13 (Climate Action)** by encouraging waste reduction, recycling, and composting actions known to lower greenhouse gas emissions [1][2].

Evaluation of the system demonstrated high usability, strong functional accuracy, and clear educational benefits. The interface was reported to be intuitive and visually engaging, and testing confirmed that the interactive features performed consistently across devices. The explanation-enhanced quiz feature played a particularly important role in deepening understanding, showing the value of well-designed feedback in environmental education [8][9].

Several opportunities for future improvement remain. Integrating real-time municipal data would increase accuracy, and enhancing accessibility features would improve support for users

with disabilities. Multilingual support, user accounts, progress tracking, and light gamification could also increase engagement while making the system more personal. Expanding the local recycling dataset would further strengthen its practical relevance. Even with these potential enhancements, the current system provides a solid foundation for scalable, community-focused environmental education.

In conclusion, WasteAware shows that simple, accessible web technologies can meaningfully support sustainable community behaviour. By bringing together interactive learning tools and practical local information, the platform encourages users to adopt environmentally responsible habits that contribute to cleaner, healthier, and more resilient urban environments advancing the goals of sustainable cities and communities.

5. References

- [1] United Nations. 2015. *Transforming Our World: The 2030 Agenda for Sustainable Development*. United Nations Publishing, New York, USA.
- [2] United Nations Environment Programme (UNEP). 2023. *Global Waste Management Outlook 2023*. UNEP, Nairobi, Kenya.
- [3] Intergovernmental Panel on Climate Change (IPCC). 2023. *Climate Change 2023: Mitigation of Climate Change*. IPCC, Geneva, Switzerland.
- [4] World Health Organization (WHO). 2022. *Health Effects of Improper Waste Management*. WHO, Geneva, Switzerland.
- [5] European Environment Agency (EEA). 2022. *Municipal Waste Management in Europe: Current Challenges and Future Opportunities*. EEA, Copenhagen, Denmark.
- [6] Ellen MacArthur Foundation. 2021. *Circular Economy and the Environment: Policy, Practice, and Impact*. Ellen MacArthur Foundation, Cowes, UK.
- [7] United Nations Human Settlements Programme (UN-Habitat). 2020. *Solid Waste Management in Cities: Improving Urban Sustainability*. UN-Habitat, Nairobi, Kenya.
- [8] Ajzen, I. 1991. The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50, 179–211. (Included as a learning-behaviour theory reference supporting the quiz and behavioural change section.)
- [9] Schunk, D. H. 2012. *Learning Theories: An Educational Perspective* (6th ed.). Pearson Education, Boston, MA. (Included to strengthen the educational justification for interactive learning and immediate feedback.)